Q1. Dataset: Retail Transactions

TID Items

T1 Milk, Bread

T2 Milk, Butter

T3 Bread, Butter

T4 Milk, Bread, Butter

T5 Milk, Eggs

Question:

Based on the dataset, explain how association rule mining can improve sales in a retail store.

Provide a practical example.

Answer:

Association rule mining helps a retail store find relationships between products that customers frequently purchase together. These patterns can be used to improve product placement, promotions, and sales.

From the given dataset:

- Milk and Bread are purchased together in T1 and T4.
- Milk and Butter are purchased together in T2 and T4.
- Bread and Butter are purchased together in T3 and T4.

For example, the rule:

Milk → Bread

shows that customers who buy Milk may also be interested in Bread.

Practical Example

A supermarket can place Bread near the Milk section or offer a discount such as "Buy Milk and get Bread at a reduced price." This can encourage customers to purchase both products, increasing the store's overall sales.

Thus, association rule mining helps retailers understand customer buying patterns, improve product placement, create targeted offers, and increase sales.

Q2. Dataset: Transaction Data

TID Items

T1 A, B

T2 B, C

T3 A, C

T4 A, B, C

T5 B, C

Question:

Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Answer:

Both Apriori and FP-Growth are used to find frequent itemsets and association rules from transaction data.

Difference between Apriori and FP-Growth

Feature	Apriori	FP-Growth
Method	Generates candidate itemsets	Builds an FP-Tree
Candidate Generation	Required	Not required
Database Scans	Multiple scans	Usually fewer scans
Speed	Slower for large datasets	Faster for large datasets
Memory	May require more memory for candidates	Requires memory for FP-Tree
Best Use	Small or simple datasets	Large and complex datasets

Using the Given Dataset

For example, Apriori first finds individual frequent items such as A, B, and C, then generates 2-itemsets such as {A,B}, {A,C}, {B,C}, and finally larger itemsets if required.

FP-Growth instead stores the transactions in a compact FP-Tree and mines frequent patterns directly without generating candidate itemsets.

When to Use

- **Apriori:** Use when the dataset is small and the algorithm needs to be simple and easy to understand.
- **FP-Growth:** Use when the dataset is large, because it avoids candidate generation and generally requires fewer database scans.

Conclusion

Apriori is simple but can be slower, while FP-Growth is more efficient for large datasets because it avoids generating candidate itemsets.

Q3. Dataset: Transaction Data

TID Items

T1 A, B

T2 A

T3 B

T4 A, B

T5 A

Question:

Explain support and confidence using the dataset and describe their importance in association rule mining.

Answer:

Support and confidence are two important measures used to evaluate association rules.

1. Support

Support tells us how frequently an item or item combination occurs in the entire dataset.

Formula:

$$Support(X \rightarrow Y) = \frac{Transactions\ containing\ X\ and\ Y}{Total\ Transactions} \times 100$$

For example, consider the rule:

A → B

A and B occur together in T1 and T4, i.e., 2 transactions out of 5.

$$Support(A, B) = \frac{2}{5} \times 100 = 40\%$$

So, the support of A → B is 40%.

2. Confidence

Confidence tells us how often Y is purchased when X is purchased.

Formula:

$$Confidence(X \rightarrow Y) = \frac{Support(X, Y)}{Support(X)} \times 100$$

A occurs in T1, T2, T4, T5 = 4 times.

Therefore:

$$Confidence(A \rightarrow B) = \frac{2}{4} \times 100 = 50\%$$

So, when customers buy A, there is a 50% chance that they also buy B.

Importance

- Support identifies item combinations that occur frequently in the dataset.
- Confidence measures the strength or reliability of an association rule.
- Together, they help businesses identify useful and reliable purchasing patterns.

Conclusion: For the rule A → B, the support is 40% and confidence is 50%. A good association rule generally has sufficiently high support and confidence.

Q4 .Dataset: Multilevel Data

TID Items

T1 Electronics, Mobile

T2 Electronics, Laptop

T3 Electronics, Mobile

T4 Electronics, Mobile

T5 Electronics, Laptop

Question:

Explain multilevel association rules using the dataset and compare them with

Multidimensional

Answer:

Multilevel association rules discover relationships at different levels of a concept hierarchy, from general categories to specific items.

In the given dataset, the hierarchy can be represented as:

Electronics → Mobile / Laptop

Example

At the higher level:

- Electronics occurs in all 5 transactions.
- Support = 5/5 = **100%**

At the lower level:

- Mobile occurs in T1, T3, T4 → 60%
- Laptop occurs in T2, T5 → 40%

Thus, multilevel mining can show patterns from the general category Electronics to specific products such as Mobile and Laptop.

Multilevel vs Multidimensional Association Rules

Multilevel	Multidimensional
Uses different levels of a concept hierarchy.	Uses different attributes or dimensions.
Example: Electronics → Mobile.	Example: Age = Young → Buys Mobile.
Focuses on general-to-specific relationships.	Combines different characteristics of data.

Conclusion: Multilevel association rules help identify patterns at different levels of detail, while multidimensional association rules discover relationships between different attributes or dimensions.

Q5. Dataset: Large Transaction Sample

TID Items

T1 A, B, C

T2 A, B

T3 B, C

T4 A, C

T5 A, B, C

Question:

Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Answer:

When the transaction dataset becomes very large, association rule mining faces several challenges.

Challenges

1. **Large number of transactions:** Processing millions of transactions requires more time and computational power.
2. **Candidate explosion:** Apriori may generate a very large number of candidate itemsets as the number of items increases.
3. **Memory usage:** Storing and processing large numbers of itemsets requires more memory.
4. **Multiple database scans:** Apriori repeatedly scans the database, which can reduce performance.

For example, in the given dataset, combinations such as {A,B}, {A,C}, {B,C}, and {A,B,C} need to be checked. In a much larger dataset, the number of possible combinations increases rapidly.

Scalable Methods

Scalable methods address these problems by:

- **FP-Growth:** Uses an FP-Tree and avoids candidate generation.
- **Parallel processing:** Divides the dataset among multiple processors or machines.
- **Database partitioning:** Processes large datasets in smaller sections.
- **Efficient data structures:** Reduces memory usage and speeds up searching.

Conclusion

Large datasets create problems such as high processing time, memory requirements, and a large number of candidate itemsets. Scalable algorithms such as FP-Growth and parallel mining reduce these problems and make association rule mining faster and more efficient.